```
Output:
int(0)
int(100)
float(100.1)
float(1)
int(1)
int(0)
int(0)
int(1111)
int(12)
```

Here we are processing an assortment of numerical strings. We're explicitly casting most of them to integers (with one example casting to a float). In order for PHP to successfully convert a string to a number, the first character encountered must be a digit, with a few caveats. Leading spaces, tabs, and the various new-line characters are fine, as demonstrated on line three with the string `"\t 100.1 mph"` (a tab character and a space prepend the numeric portion of the string). PHP will continue to convert the string until it hits a non-digit character, the exception to this rule being the decimal point: if PHP is attempting to cast to a float, then a single decimal point is included along with all digits immediately following. If casting to an integer, a decimal point will end the conversion.

Thus, `'speed: 100.1 mph` casts to *0* (the value for any non-numerical string), `'100.1'` casts to (integer) *100* or (float) *100.1*, and `'1,000,000'` casts to *1* (numeric type casting is not aware of number formatting). The remaining examples demonstrate attempts to cast numbers from different base systems (hexadecimal, binary, and octal). In all cases the conversion rules remain the same, and they are treated as Base 10 in PHP 8 (although PHP was actually capable of converting hexadecimal strings until version 7.0, this "feature" was removed).

At the beginning of the section I mentioned that numerical type juggling could lead to security problems; let's look at one. Imagine we have a secret stored somewhere, a hexadecimal string generated from a hashing algorithm like SHA256. We compare this to a user-supplied token, which is submitted to our application as a JSON data structure; if they match, then access is granted. What could possibly go wrong if that secret hash string began with a `'0'`? Quite a bit, as it turns out.

```
$secretHexValue = 'b94d27b9934d3e08a52e52d7da7dabfac484efe37
a5380ee9088f7ace2efcde9';
$userInput = '{ "token": 0 }';
```